



Cobot Flippy at Dodger Stadium, Los Angeles, USA (Courtesy: Miso Robotics)

and so on. Universal Robots, ABB and Fanuc are some of the leading international players in this almost-mainstream technology, while Systemantics is a completely indigenous maker of collaborative robots.

Although not developing as fast as cobots, autonomous delivery is another interesting tech to watch this year. Autonomous delivery systems are small, electric-powered motorised vehicles, which can deliver goods to people autonomously. There have been trials using four-wheeled robots and drones. They are generally lightweight with soft, rounded edges, and all the works to ensure that they cause no harm even if they accidentally bump into humans or animals. They will carry packages to be delivered within digitally-locked containers. A user must enter a one-time-password or some other means of authentication to open the container and collect the package.

We have been reading about autonomous cars and trucks for a long time now but, honestly, it is not clear when we will really see one on the roads of an Indian city! However, as a spin-off of the research in artificial intelligence, sensors, Internet of Things (IoT), lidar, and other component technologies, as well as fundamental research in applied math and computer science, we can

expect to see autonomous delivery systems in the near future. Initially, we can only expect to see localised autonomous delivery systems, perhaps within a factory or a university campus. There have been some initial setbacks, as we can deduce from the likes of Amazon and FedEx scaling back their autonomous delivery programs. But several companies are still working on the tech—such as Starship Technologies, Nuro, and Robby Technologies—and market reports suggest that the industry is still on a growth trajectory. According to a report by Fortune Business Insights, the global autonomous last mile delivery market size is projected to grow from \$11.12 billion in 2021 to \$51.38 billion in 2028, at a compound annual growth rate (CAGR) of 24.4% during forecast period. One of the key drivers for pursuing this technology is the need felt for such systems during the Covid pandemic.

Cobots and autonomous delivery systems are interesting technologies to watch, because they work alongside humans, with no intention to replace them! This is reason enough for them to succeed.

Ubiquitous AI

Artificial intelligence is in a really advanced state now. We know it

sits in the highest places—from airplanes and race cars to medical equipment and robots. But, according to a Forbes trendwatcher, the interesting trend we are going to see in 2023 is ‘AI Everywhere.’ With the emergence of no-code AI platforms and with boundless computing power available as-a-service, more and more people are beginning to use AI to run their businesses better. Earlier, only AI experts could develop AI applications, but now domain experts can do it themselves without any complex code. Some apps like Sway.ai use a drag-and-drop interface, while others like Akkio use a wizard based one.

Some of the trending no-code AI apps today are MonkeyLearn, Levity AI, obviously.ai, sway.ai, Akkio, NoCode.ai, Google AutoML, Juji, Lobe.ai, and Microsoft Power Platform, among others. Businesses across the world are using no-code AI platforms in interesting ways—such as for watching bees in farms to prevent deadly pest attacks, in autonomous shopping systems, and more.

Web3

Web3, long touted to be a better version of the internet, is slowly coming of age, as we see many of its components coming into mainstream use. According to Ethereum, where the concept originated in 2014, “Web3 has become a catch-all term for the vision of a new, better internet. At its core, Web3 uses blockchains, cryptocurrencies, and NFTs to give power back to the users in the form of ownership. A 2020 post on Twitter said it best: Web1 was read-only, Web2 is read-write, Web3 will be read-write-own.” It centres around the concept of decentralisation, giving ownership to users, rather than power being concentrated in the hands of a few powerful organisations.